Advanced Time Series Analysis: From probabilistic to foundational models

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ABSTRACT

Time series analysis is fundamental in many fields such as finance, health care, climate science, autonomous systems. This seminar discusses about three broad categories of time series model:explicit,implicit and task agnostic and foundational models.Explicit models include Interleaved Hidden Markov Model, Flow HMMs, Wavenet and DeepAR. These are probabilistic models and have strong assumptions about the data generation and have domain specific knowledge,these models also focus on the temporal dependencies explicitly. The implicit model includes BRITS and various Transformer based architectures (PatchTST, iTransformer) that focuses on temporal dependencies directly from data without any explicit assumption. The task and foundational models such as BERT, MOIRAI, Visual Autoregressive Models, and Latent Diffusion Models highlight the potential to be generalized well to other domains. Through comparative analysis of various models we analyze the trade offs in scalability , interpretability, etc. and model selection based on the task requirement and dataset characteristics.

Keywords Time Series · Probabilistic Time Series · Transformers · Foundational Models

1 Introduction

In the field of Machine Learning, Time-Series Analysis and Prediction is a very important concept that has been used in various real-life applications. It helps in making informed decisions, optimizing outcomes, mitigates risk, planning, anomaly detection and imputations in various domains such as businesses, healthcare, financing, climate, resource management, etc. A time series is a set of observations arranged chronologically. A times series model is a machine learning model which analyzes the sequential data in the dataset and predicts the future values Parzen [1961]. The time series has four main components as shown in Table 1. One of the key assumption made by time series is the stationarity of the dataset that means the variance and mean is constant over time. The time series can be divided into two types based on the features as **uninvariate** : time series with one variable observed over time and **multivariate** time series: time series with two or more variables over time.
In this paper, we discuss the different approaches to time series modeling. We have divided it into mainly 3 types: (1) explicit time modeling (2) implicit time modeling (3) task-agnostic approaches. We analyze different models under each approach.In the explicit models we discuss about the conventional probabilistic techniques used in the time series. In the implicit models we discuss about the models based on neural networks and transformers. Last, we discuss about task agnostic where we see how different foundational models can be used under the context of the time series.

2 Explicit Modeling of Time

An explicit model is one that encodes assumptions about data generation process, where the distribution assumptions are specified. Like in interleaved Markov model Landwehr [2008] it explicitly define the transition, emission and switching process for the probabilistic modeling sequence.Whereas in FlowHMMs Jo et al. [2022] it models transition

Component	Description
Seasonality	Repeating patterns or cycles at regular intervals (e.g., annually, monthly).
Trend	Movement in the data showing overall increase or decrease over time.
Cyclic	Fluctuations around the true data distributions occurring at irregular intervals.
Noise	Random variation in the data that cannot be explained by the other components.

Table 1: Main Components of a Time Series

between time steps explicitly and adds emission modeling.In DeepAR Salinas et al. [2017] though it uses the RNNs but still it defines an output distribution explicitly. We consider Wavenet van den Oord et al. [2016] explict as it models the raw sequences explicitly.

2.1 Modeling Interleaved Hidden Processes

This paper Landwehr [2008] discusses about a novel probabilistic framework for modeling scenarios where multiple independent processes generate a single observed sequence. The key innovation were:
1) Switching process
2) Frozen inactive process
3) Tractable approximate inference via chainwise Viterbi algorithm

2.1.1 Method

- Hidden Process:
  States: $s^{(m)}$, transitions: $a_{ij}^{(m)}$, emissions: $b_i^{(m)}$
- Switching Process:
  Markov chain over $Z_t \in \{1, \ldots, M\}$ with transitions a_{ij}
- Observations: Y_t

Key Equation

$$P(Z, S, Y) = \prod_{t=1}^{T} P(Z_t \mid Z_{t-1}) \cdot P(Y_t \mid S_t, Z_t) \cdot \prod_{m=1}^{M} P(s_1^{(m)}) \prod_{t=2}^{T} P(s_t^{(m)} \mid s_{t-1}^{(m)}) \tag{1}$$

For the inference the goal is to Estimate $\hat{Z}, \hat{s}^*$

$$\arg\max_{Z} \arg\max_{s} P(Z, s \mid Y) \tag{2}$$

The inference has few challenges. This is a NP-hard problem so it is dificult to compute and it has temporal dependencies. Therefore, the authors used the chainwise viterbi algorithm.

$$S_{ijt}^{(m)} = \max P(history\ s_t^{(m)} = i,\ s_{t+1}^{(m)} = j,\ Z_t = z_k) \tag{3}$$

The complexity is:$O(K^2 M(N+K)T)$ which is less the the HMM and it also reduces the parameters.

2.1.2 Results and Discussion

The experiments were performed on dataset of 20 sequences and 5 persons. The HMM_{mix} performed better than the flat HMM. Thus, it introduced a model for interleaved mixtures of hidden processes, outperforming standard single-process HMMs in activity recognition but it is applicable in scenarios where only the interleaved output of multiple processes is observable. The **scalability** of large M and K is still a challenge for the model.The switching and frozen inactive processes are not evolved unless active which is difficult in a real world scenario.

Thus,Interleaved Markov is an explict time model as it explicitly define the transition, emission and switching process for the probabilistic modeling sequence.The assumptions about Markov properties,stationarity are clearly stated.

2.2 FlowHMM: Flow-based continuous hidden Markov Models

The continuous hidden Markov models assume that observations are generated from a mixture of gaussian densities which limits their ability to model complex distributions. The traditional HMM also uses EM(Expectation Maximization) for the training which is very less efficient and slow convergence. This paper Jo et al. [2022] provides a novel method that enable learning general continuous observation densities without constraining them to follow a Gaussian distribution or their mixtures. The FlowHMM are divided into two variants. The first is based on the gradient based technique and the second one relies on the co-occurence matrix of discretized observations and considers the joint distribution of pairs of co-observed values.

2.2.1 Method

The HMM structure remains the same but the emission likelihoods are computed using Continuous Normalizing Flows(CNFs) instead of Gaussian.The two proposed variants of FlowHMM models differ mainly in training process.
Training FlowHMM F^{ML}
FML is trained with an maximum likelihood approach directly on continuous data via gradient methods. There are no discretizations applied. They are computationally heavy for long sequences because each epoch processes all data through the forward algorithm.
Training FlowHMM F^{Q}
FQ utilizes co-occurrence matrix and is trained in end-to-end setting using discretized sequences. It discretizes the observations on a grid and then use the co-occurence matrix ad learns the model parameters to match it using the KL divergence.

2.2.2 Results and Discussion

From the experiments we see that in synthetic 1D data for small sequence length F^{ML} performs better than F^{Q}.In Real 1D data F^{Q} outperforms F^{ML} . Whereas in 2D synthetic data F^{ML} outperforms F^{Q}. In the metrological dataset of 6D F^{ML} outperforms the gaussian HMMs.
It assumes for stationary process only. F^{Q} heavily depends on the co-occurence matrix. F^{Q} also struggles with high dimensional data.

We can conclude that F^{ML} are better for high dimensional sequences and F^{Q} for low dimensional sequences.FlowHMM, consistently performs better than gaussian HMM for complex,non-gaussian emission distributions. We consider FlowHMM to be explicit as it parametrizes the emission distribution with a normalizing flow rather than assuming it to be a gaussian. Thus, making it explicit time modeling.

2.3 WaveNet: A Generative Model for Raw Audio

This paper van den Oord et al. [2016] talks about auto-regressive model for generating high-fidelity raw audio waveforms. The paper contributes by modeling each audio sample conditioned on all previous samples,capturing long term dependencies. It also uses dilated causal convolutions which enables exponential growth of the receptive fields without excessive computation. It also introduces conditioning which captures global and local dependencies.

2.3.1 Method

Wavenet introduce a new generative model operating directly on raw audio waveform. The joint probability of a waveform $\mathbf{x} = x_1, \ldots, x_T$ is factorized as a product of conditional probabilities. Each audio sample x_t is therefore conditioned on the samples at all previous timesteps. The main ingredient is the causal convolutions which ensures that the future timesteps does not influence the past and it follows the temporal ordering. Authors use dilated convolutions for large receptive fields to capture long range dependencies.For fast and deep network training residual and skip connections are used. Then each sample value x_t is quantized to 256 values using the μ law. The output layer predicts the softmax distribution over these quantized levels.We condition them globally or locally by adding an additional input h given in equation(4). By conditioning the model on other input variables, we guide WaveNet's generation to produce audio with the required characteristics.So, the global conditioning is characterised by a single latent representation h that influences the output distribution across all timesteps and the local conditioning we have a second timeseries h_t, possibly with a lower sampling frequency than the audio signal.The loss is calculated using the cross entropy between the softmax distribution and actual quantized audio sample.

$$p(\mathbf{x}) = \prod_{t=1}^{T} p\left(x_t \mid x_1, \ldots, x_{t-1}, h\right) \tag{4}$$

2.3.2 Result and Discussion

In the Text to Speech experiment wavenet outperforms LSTM-RNN and HMM.
The main **limitation** of the wavenet is the limited long range coherence. The receptive field hinder the global structure. It also requires a large dataset and have a high computational cost.

Thus, due to prediction of each sample conditioned on past samples enforces explicit temporal dependencies making the Wavenet an explicit time modeling approach. It does not assume stationarity and works directly with raw audios.It is very arguable to consider the Wavenet as an implicit model also because it encodes the knowledge and captures the local and global features implicitly.

2.4 DeepAR: Probabilistic forecasting with autoregressive recurrent networks

It is a forecasting method Salinas et al. [2017] based on autoregressive recurrent neural network(RNN), designed to handle large scale time series data. Traditional model struggles with the scalability and model selection. DeepAR employs an autoregressive RNN architecture with LSTM cells to model the conditional distribution of future time series.It mitigates the issues regarding error metrics skewed by scaling, weighted sampling and using flexible distributions.

2.4.1 Method

Each time step model takes the previous observation and the covariates $x_{i,t}$ and previous hidden state $h_{i,t-1}$ The key features of the DeepAR model are the following:
1)**Likelihood Models**: Gaussian for real-valued data,negative binomial for count data and bernoulli for binary data.
2)**Scale Handling** : Time series vary in magnitude so DeepAR addresses power law distribution by normalizing the samples and high magnitude series are trained more frequently according to the weighted sampling .
3)**Covariates**: It incorporates time dependent and item dependent features.
4) After incorporating all the features the model is trained on multiple instances, that is created by the sliding window and tries to optimize log likelihood maximization. The DeepAR does not forecast pointwise prediction, instead it generates Monte Carlo samples,these samples are estimated as quantiles for decision making under uncertainity.

2.4.2 Results and Discussion

DeepAR outperforms all the baseline models as discussed in the paper especially large time series dataset. Thus, DeepAR capability to learn seasonal behaviors and dependencies on given covariates across time series is needed in order to capture complex, group dependent behavior. Also,by learning from similar items, DeepAR is able to provide forecasts for items that have little or no history available but DeepAR **lacks interpretability** and the performance on highly irregular or sparse time series is untested.

DeepAR is an explicit time series model as it explicitly models the likelihood function and also it explicitly depends on the past data for the prediction of future observations by decomposing the joint probability to product of conditional distribution.

3 Implicit Modeling of Time

An implicit model is one where the relationship between the variables or the function is not directly stated. In this section all the four papers uses neural network or transformers which learns the temporal dependencies from the data and predict or impute the values. In BRITS Cao et al. [2018] they use bidirectional recurrent model whereas in iTransformers Liu et al. [2024] and A Time series is Worth 64 words Zeng et al. [2023] they used transformers to map the temporal dependencies using large dataset. No mathematical formulation or assumption was explicitly defined.

3.1 BRITS: Bidirectional Recurrent Imputation for Time Series

Time series data is used in almost every classification and regression applications (eg- healthcare, activity classification). In real-life applications time series data has many missing values. This paper Cao et al. [2018] discusses about the imputation of the time series data and then predicting the values. The paper also discusses about the previous models that are used for imputation but they have some limitations. There are strong assumptions about linearity in ARMA models or treated imputed values as constant in M-RNN and some models ignore feature correlations like GRU-D model. In BRITS they directly uses the bidirectional RNN for the imputation of missing values. Hence does not require any specific assumption.It regards the missing values as variables and simultaneously performs the regression/classification.

3.1.1 Method

The BRITS help the model to converge fast and does not have delayed error which causes bias exploding. It utilizes the bidirectional recurrent dynamics and goes in both forward and backward direction. Thus the model compute the backpropagation error in both directions. The BRITS enforce the prediction in each step to be consistent in both directions by introducing the consistency loss equation(5).

$$l_t^{\text{cons}} = \text{Discrepancy}(x_t, \hat{x}_t) \tag{5}$$

The final estimation value is the mean absolute of $(x_t, \hat{x}_t)$. Here we consider that all the features are uncorrelated.
In correlated imputations they introduce a feature based variable for the estimation. We combine the weights carried by the feature and the history based estimation.Thus, we learn the weight by considering both the temporal decay and the masking vector m_t . The rest is similar as uncorrelated case.

3.1.2 Result and Discussion

The BRITS perform the best among the non-RNN and RNN models when performed for the imputation task on the air quality dataset. For classification tasks BRITS also has better AUC than M-RNN and GRU-D. There are still some **limitations** of the BRITS model.The time gap can sometimes be misleading. The feature based imputation assumes fixed linear relationships but in real life data it is non-linear.In long sequences gradients may explode or vanish.
BRITS learns the missing values in bidirectional recurrent system without any specific assumptions. Thus, an implicit time modeling approach.

3.2 Attention Is All You Need

In this paper Vaswani et al. [2017] an attention layer was introduced in the encoder decoder sequential models to compute the representations of its input and output. It allows for efficient parallel computation and improves models long range dependencies compared to RNNs,LSTMs or GRUs. They require large data for training. The transformers were built for the NLP tasks so they treat each token as discrete but the times series data is continuous. Therefore, some modification needs to be done or adding task specific head in this case for forecasting or imputation for the transformer that we will discuss further in iTransformers and Time series is Worth 64 words.

The reason we consider them in implicit is because of the self-attention layer and the Feed Forward Network as no specific assumption is made as it can deal with any type of data and does not assume stationarity but the positional encoding for mapping of feature embeddings is explicit as it explicitly gives relative positions to the tokens. Next sections we will see how the transformers can be used for time series modeling.

3.3 A Time Series is Worth 64 Words: Long-term Forecasting with Transformers

This paper Zeng et al. [2023] tried to address the problem of using each time step as single token by the transformers. They also address the problem of long term forecasting as they struggle due to quadratic complexity and inefficiency to capture the temporal dependencies. The paper introduces patching that segments into subseries-level patches which serve as input tokens to Transformers and captures semantic information enables longer look back window. The PatchTST is also channel independent so each univariate time series is processed independently but share same transformer weights.

3.3.1 Method

The time series is segmented into fixed patches and then generate specific attention maps for the channel independence. Each univariate is processed independently through the same Transformer backbone to avoid feature mixing. Positional encoding is used to maintain temporal ordering. Masked autoencoder are used to randomly mask patches and reconstruct them. This enables pretraining and transfer learning.

3.3.2 Result and Discussion

PatchTST outperforms FEDformer,DLinear,Autoformer across 8 benchmark datasets(Weather,Traffic,Electricity, etc.) especially for large datasets.
PatchTST handles longer look back window efficiently and captures local temporal effectively. The **drawbacks** of PatchTST is that the patch size is fixed thus may not adapt to different temporal settings and it does not perform well on the small dataset.The channels are independent thus ignore the cross channel dependencies.
Thus,learning patterns directly from the data and no explicit likelihood formulation makes Patch TST an implicit approach.

3.4 iTransformer: Inverted Transformers Are Effective for Time Series Forecasting

This paper Liu et al. [2024] discuss about the approach to adapt transformers for multivariate time series forecasting. It is an encoder only architecture. The major challeneges with the traditional transformers are that they loss of the variates specific information. The variates with different physical meanings or measurements are fused into indistinguishable channels. They have meaningless attention maps as time aligned point may not represent the same event. This paper proposes variate tokens by embedding entire time series of each variate independently that is why inverted transformers. This helps to capture the multivariate correlations across the variates using the attention maps. Feedforward network process the temporal dependencies for variates implicitly.

3.4.1 Method

Instead of treating each timestamp as token, model treat each variate as entire time series as tokens. Then in the encoder only architecture it apply self-attention on variate tokens to model cross variate dependencies. Then FFN captures the patterns (e.g. periodicity, trend etc.). Layer normalization is applied per variate to normalize representation and reduce discrepancies. For training supervised learning is used on long term forecasting tasks using MSE loss.

The iTransformer ourperforms state of the art models FEDformer,PatchTST on benchmarks as ETT,Weather,Traffic and electricity. The reason iTransformer performs better than PatchTST is that it handles the cross feature dependencies. It is effective on high dimensional datasets. One disadvantage is that they are computationally expensive for extremely large number of features.

4 Foundation Models and Task-Agnostic Approaches

Task agnostic approaches are the general purpose representation of the model that can be applied for various downstream tasks. In this section we will see general purpose models such as BERT, MOIRAI, VAR and Latent diffusion models and how relevant they are in context of time series and can learn the general purpose temporal representations.

4.1 BERT:Pre-training of Deep Bidirectional Transformers for Language Understanging

BERT: Bidirectional Encoder Representations from Transformers Devlin et al. [2019] uses masked language models to enable pre-trained deep bidirectional representations. We also use a next sentence prediction task that jointly pre-trains text-pair representations. It is an encoder only model.In the Masked Language Model(MLM) pre-training it uses 15 percent masking. MLM helps in bidirectional understanding of words. The NSP helps in understanding relationship between sentences.
Since, the BERT pre-trains on unlabeled dataset and embeds the general representations of the text and can be used for various downstream tasks, this makes BERT task agnostic approach.

4.1.1 Method

There are two main steps in BERT: pre-training and fine-tuning. During pre-training, the model is trained on unlabeled data over different pre-training tasks. For fine-tuning, the BERT model is first initialized with the pre-trained parameters, and all of the parameters are fine-tuned using labeled data from the downstream tasks such as natural language inference, question answering, named entity recognition ans sentinment analysis grammar checker. These are evaluated on the metrics GLUE(General Language Understanding Evaluation) benchmark.

4.1.2 Discussion

In context of timeseries there is a TimesBERT model Zhang et al. [2025] which is similar to the BERT. It considers the sentence as time series and the tokens as time steps. It randomly mask the time series and train the model to reconstruct the missing values. This pre-training learns the general temporal representations. This can be used for various downstream tasks such as forecasting, missing value imputation etc. TimesBERT Chen et al. [2023] was proposed for financial fraud detetecion and stock selection task where it performed better than the LSTMs.

4.2 Unified Training of Universal Time Series Forecasting Transformers (MOIRAI)

It Woo et al. [2024] is a novel Transformer based designed for universal time series forecasting. However, constructing such a model poses unique challenges specific to time series data: a) Cross frequency learning b) any variate forecasting

c) varying distribution. It is trained on Large-scale Open Time Series Archive (LOTSA) featuring over 27B observations across nine domains.

4.2.1 Method

A masked encoder based Transformer with modifications to handle heterogeneous time series data.

1)**Patch based Input**: This divides high frequency data into large patches to reduce the computational cost and a smaller patch size for low-frequency data.

2)**Any Variate Attention**: It uses Rotatory Positional Embedding for time and binary attention biases for variate disambiguation.

3)**Mixed Distribution**: This predicts parameters for a mixture of distribution that includes student t-distribution, negative binomial for positive count,log normal distribution to right skewed distribution and low variance gaussian distribution.

4)Given the Equation(6),they pre-train task to optimize the mixture distribution log-likelihood.

$$\max_{\theta} \mathbb{E}_{(Y,Z)\sim p(D)} \mathbb{E}_{(t,l,h)\sim p(T|D)} \log p(Y_{t:t+h} \mid \hat{\phi}), \quad \hat{\phi} = f_{\theta}(Y_{t-l:t}, Z_{t-l:t+h}) \tag{6}$$

where $p(D)$ is the data distribution which samples for a time series (Y, Z), and $p(T \mid D)$ is the task distribution which defines the lookback window, $Y_{t-l:t} = (y_{t-l}, \ldots, y_{t-1})$ with context length l and forecast horizon, $Y_{t:t+h} = (y_t, \ldots, y_{t+h-1})$ with prediction length h.

MOIRAI is capable of probabilistic and long sequence forecasting.They show that as a zero shot forecaster,MOIRAI achieves competitive or superior performance compared to full-shot models.They also introduced LOTSA largest open data for pre-training time series forecasting model. One **limitation** of the model is that the current architecture has limited support for high-dimensional time series.

4.3 Visual Autoregressive Modeling: Scalable Image Generation via Next-Scale Prediction

In the traditional auto-regressive models images are converted to discrete tokens, then tokens are flattened to form 1D sequence and a transformer predicts tokens one by one in this sequence. They are very inefficient($O(n^6)$ and poor scalability images. The VAR Tian et al. [2024] addresses this problem by redefining the next token to next scale (coarse to fine).

4.3.1 Method

The intuition behind the Next Scale Prediction is that we start with a coarse scale and then progressively predict finer resolution (4x4,8x8 etc.). Full token map is predicted in parallel,conditioned on previous coarser maps. The first in the architecture is tokenization. It is done by multi-scale Vector Quantized Variational Autoencoder (VQVAE).It encodes an image into multiple resolution token maps $r_1, r_2, ...r_k$. The shared book across the scales ensures the consistency. Then it is trained with the VAR Transformer. It trains to predict each scale r_k conditioned on $r_1, ...r_{k-1}$. The efficiency is reduced to $O(n^4)$. Also, VAR had the lowest FID and highest Inception score(IS) among the GAN,diffusion and AR models.

4.3.2 Discussion

They also discuss about the scaling law and how it improves the test loss as the Autoregressive model scale up. The scaling improves the zero shot generalization.The limitation of the VAR is that it is dependent on the quality of the VQVAE. Also, the text to image is not yet implemented.
In the context of time series and task agnostic approach the VAR performs tasks such as inpainting which can be looked as interpolation in time series and outpainting which is like forecasting that is predicting the what comes after. It is breakthrough in AR model for vision. It is also foundation model for zero shot generalization.

4.4 Align Your Latents: High-Resolution Video Synthesis with Latent Diffusion Model

Latent Diffusion Models (LDMs) Blattmann et al. [2023] enable high-quality image synthesis while avoiding excessive compute demands by training a diffusion model in a compressed lower dimensional latent space. In this paper the authors apply the LDM paradigm to high-resolution video generation. This can be used in task such as autonomous

driving and text to video or creative content creation. The authors add a temporal layer to turn the image generator to video generator. Video LDM model time implicitly through temporal layers.

4.4.1 Method

First the image LDM is pre-trained on large image datasets. Then temporal attention layers is added.They are trained only on the temporal layers to learn time consistency. Then it go to the temporal decoder fine tuning to ensure smooth video output.This gives high resolution long videos. While enhancing the video the model adds frames between keyframes for smooth motion. It uses the upsampler to bring the original resolution or boost the resolution of the video.

4.4.2 Result and Discussion

The model was evaluated on tasks driving video synthesis, text to video and personalized text to video. FVD (Frechet Video Diatance) and IS was used for the evaluation metrics. It achieved better results than Long video GAN on high driving video generation. In the text to video generation also Video LDM performs better than Make-a -Video. In context of time series real application it was used by Rasul et al. ,Rasul et al. [2023] they used LDMs to synthesize realistic patient trajectories.

The Video LDM generalizes across tasks like autonomous driving and text to video.It is implicit also as the temporal attention and 3D convolutions learn frame relationships without explicit time steps. The main drawback of the Video LDM are that they are not yet indistinguishable from real content.

5 Conclusion and Future Direction

Explicit Models Thus, these models are **interpretable** and they have **strong assumptions**. HMMs are still state of the art in many domains. Interleaved HMMs and Flow HMMs work well for small datasets with strong domain knowledge. Deep AR is effective for medium size related time series with covariates. If we want to model the high resolution waveforms then Wavenet is an effective model. However, **scaling** explicit models and improving interpretability in complex settings remain challenging.

Implicit Models Models that does not have any assumptions,handling non linear and high dimensional relationships and learns the temporal patterns directly from the data are implicit models. BRITS is robust to the missing data and imputes sparse or irregular data.Patch TST captures long range dependencies in univariate time series,iTransformers are used for multivariate time series forecasting across variates.The limitations of transformer based models are they are data guzzlers and computationally expensive.Also, struggles with **modeling temporal structures** explicitly and **weaker interpretability**. Models that captures the temporal structure like time aware attention is a challenge for transformers and a future research work.

Foundation Models These are trained for general purpose representation and can be applied to various downstream tasks. Like TimesBERT for masked time series modeling. This can be used for classification and imputations of time series dataset. MOIRAI for zero shot or few shot forecasting. We also look at the vision models like VARs(Visual Autoregressive Model) which can be used for sequence generation. Latent difusion models incorporates temporal dependencies to synthesis high resolution and smooth videos. Main challenges of these models are **domain generalization** and **adapting vision/language architectures** to time series.

Future Direction Hybrid architectures combining explicit probabilistic models (e.g., HMMs) with implicit neural networks (e.g., transformers) are emerging to bridge interpretability and scalability. HybridNets Smith and Doe [2023] integrate HMMs with attention mechanisms for healthcare time series. However, challenges like alignment between modules and training stability remain open.

Model choice depends on data availability, domain knowledge and task requirements. The practical approach is to start with simpler models and progress to more complex as needed.

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